

Probabilistic project scheduling with SPERT® three-point estimation and Monte Carlo simulation

WHAT IS SPERT® SCHEDULER?

SPERT® Scheduler is a lightweight, probabilistic scheduling tool for project managers working on traditionally-scheduled projects. It combines Statistical PERT three-point estimation with Monte Carlo simulation to answer: *"How confident are we that we finish by a given date?"* — and to calculate the schedule buffer needed. It is independent of SPERT® Forecaster (agile tool).

Three-point estimates	Enter Optimistic (min), Most Likely, and Pessimistic (max) durations in working days per activity. Enable Heuristic Mode in Settings → Preferences to auto-calculate Min and Max as configurable percentages of Most Likely (e.g., Min = 75%, Max = 200%).
Confidence level	Applies only to T-Normal and LogNormal distributions. Maps your subjective confidence (Near certainty → Guesstimate) to a statistical standard deviation via the SPERT Ratio Scale Modifier (RSM). Has no effect for Triangular or Uniform distributions.
Distribution types	T-Normal, LogNormal, Triangular (default), Uniform. An automatic recommendation is shown based on your estimate spread. Confidence level is not applicable to Triangular or Uniform.
Deterministic schedule	Each activity's scheduled duration = $\text{inverseCDF}(\text{activity probability target, e.g., P50})$. The schedule is calendar-aware, respecting your configured work week and all holiday layers.
Monte Carlo simulation	Default 10,000 trials per run; up to 100,000 trials available for audit-grade precision where reduced variance is needed. Each trial samples a duration for every activity from its distribution. Parkinson's Law clamping (configurable per scenario): no simulated duration falls below the deterministic scheduled duration.
Schedule buffer	Buffer = MC project percentile at your project target – deterministic total duration. Set a Finish Target date to see RAG (red/amber/green) schedule health in the summary card and on the Gantt chart. Thresholds configurable in Settings (defaults: Green = P80, Amber = P50).

■ **Tip:** Use T-Normal or LogNormal when you have a confident Most Likely estimate and want the Confidence level to influence the spread. Use Triangular or Uniform when you prefer a simpler, assumption-free distribution.

STEP-BY-STEP WORKFLOW

#	Where	What to do
1	Projects page	Create a new project (tile-based list) or open an existing one. Drag tiles to reorder. Search or archive projects for tidy housekeeping. Assign a tile color (Slate, Sage, Sky, Lavender, Rose, Amber, Teal, Clay) to color-code projects by program — shown as a 4px accent strip on the tile. Clone any project from its tile to produce a fully-detached copy named '{original} (Copy)' — useful for scrubbing sensitive data before sharing as a reproducer without touching the original. Use Export All Projects or Import Projects from the page header for bulk backup and restore. A theme toggle (Light / Dark / System) in the app header lets you switch color modes globally.
2	Calendar & holidays	Configure your work week (any combination of days), add country public holidays via the Holiday Loader (100+ countries), and enter manual holidays. All scheduling respects the active calendar.
3	Add activities	Type activity names directly into the grid, or import in bulk: Settings → Import Activities from Spreadsheet accepts a CSV file or clipboard paste with a downloadable template. Use the + Section button (beside + Add Activity) to insert section header rows — each with an editable name and optional color — to visually group activities in the grid and Gantt chart.

4	Estimate & configure	Enter Min / Most Likely / Max per activity (or let Heuristic Mode auto-fill). Set Confidence level (T-Normal / LogNormal only) and Distribution. Use Settings → Estimation Heuristics Suggester for evidence-based min/max percentage defaults across 23 industries and 73 activity subdomains.
5	Dependencies & targets	Toggle Dependency Mode to add Finish-to-Start / Start-to-Start / Finish-to-Finish relationships with optional lag, apply scheduling constraints (MSO, MFO, SNET, SNLT, FNET, FNLT), and view CPM critical path and float analysis. Set a project-level Finish Target date in the summary card for RAG schedule health tracking.
6	Simulate, report & export	Click Run Simulation. Review the histogram, CDF chart, and percentile table (P5–P99). The RAG indicator reflects schedule health vs. your Finish Target. Enable Fit to Window in the Gantt appearance panel for multi-year programs — the full timeline fits in one image for PowerPoint or Outlook. Compare scenarios side-by-side, run Sensitivity Analysis, and export the schedule as XLSX or CSV.

Note: Simulation runs automatically when you change estimates or distributions. Re-run manually after adding activities, changing dependencies, or updating scheduling constraints to refresh all results.

ACTIVITY GRID — COLUMN GUIDE

Name	Activity name. Click to rename inline. Hover any row to reveal a pencil icon that opens the Activity Edit modal directly. The # column shows sequential numbers when Show Activity IDs is enabled (Gantt toolbar — persists per project). Status icon prefix: circle = Planned, filled square = In Progress, checkmark = Complete.
Min / ML / Max	Three-point estimates in working days. Heuristic Mode auto-fills Min and Max from Most Likely using your configured percentages. All three fields are editable directly in the grid.
Confidence	Subjective confidence label (Near certainty → Guesstimate). Applies to T-Normal and LogNormal only. Blank for Triangular and Uniform in exports.
Distribution	T-Normal, LogNormal, Triangular, or Uniform. A sparkline preview and auto-recommendation are shown. Changing distribution with heuristics on prompts you to recalculate Min/Max.
Scheduled	Calculated start and end dates for this activity based on the deterministic schedule. Respects work week, holidays, converted work days, and constraints.
Duration	Deterministic scheduled duration = inverseCDF(activity target, e.g., P50). Shown in working days.
Status / Actual	Set activity status: Planned, In Progress, or Complete. Actual Duration is editable for In Progress (elapsed working days from scheduled start — auto-filled on status change, editable to override) and for Complete. In-progress elapsed days floor the simulation: no trial can be shorter than elapsed + 1 day. Bulk-mark Complete auto-fills actual durations from elapsed days.
Task progress bar	Thin bar beneath the activity name (blue = tasks in progress, green = all tasks complete). Click to open the Activity Edit modal → Tasks section. Up to 50 tasks per activity.
Deliverables bar	Second thin bar (indigo = deliverables in progress, green = all delivered). Shown alongside the task bar when both are present. Click to open the Activity Edit modal → Deliverables section. Up to 50 deliverables per activity.
Notes indicator	A small violet bar beneath the activity name (alongside task and deliverable bars) indicates the activity has notes. Click to open the Activity Edit modal → Notes section.

Constraint	Dependency Mode only. Clickable badge showing the constraint type (MSO, SNET, etc.) and Hard/Soft mode. Opens the Scheduling Constraint section of the Activity Edit modal.
Section headers	Bold divider rows inserted via the + Section button (next to + Add Activity). Each section has an editable name, an optional color (applied as a faint tint to the section header row itself), and a delete button. Up to 50 per scenario; locked scenarios prevent changes. Carry no scheduling logic and do not affect simulation results.

ACTIVITY EDIT MODAL

Open by clicking a Gantt bar, the task or deliverables progress bar, the constraint badge in the grid, or by hovering any activity row and clicking the pencil icon. The modal has up to eight collapsible sections. Estimates section is collapsed by default. Blue dot on section headers (Scheduling Constraint, Dependencies) and violet dot on the Notes section header indicate content is present. Unsaved changes guard: clicking outside the modal with pending changes prompts you to save or discard.

1 • General	Activity name and status (Planned / In Progress / Complete). When a schedule has been computed, a second row displays: Sched. Start , Sched. Finish , and Sched. Dur. (all read-only) alongside Actual Dur. (editable for In Progress and Complete) and Actual Finish Date (editable for Complete only). Entering either actual field auto-calculates the other using the project work calendar — working-day math accounts for holidays and the configured work week. For In Progress, Actual Duration is pre-filled with elapsed working days from the scheduled start when you change status; it remains editable to override.
2 • Estimates	Min, Most Likely, Max, Confidence level, Distribution type, and Activity probability target. Heuristic Mode auto-recalculates Min/Max when ML changes.
3 • Scheduling Constraint	Select constraint type (MSO, MFO, SNET, SNLT, FNET, FNLT), date (non-working days snap forward), Hard or Soft mode, and an optional note (≤ 500 chars). Live conflict preview shown. Blue dot on section header when a constraint is set.
4 • Dependencies	Dependency Mode only. Add predecessors/successors directly in the modal. Choose relationship type (Finish-to-Start / Start-to-Start / Finish-to-Finish) and optional lag days. Negative lag (lead time) is supported — enter a value like -5 to overlap activities. Lag values are clamped to ± 365 days on save. Blue dot when dependencies exist.
5 • Tasks	Add, toggle, reorder (drag), and delete task checklist items. Up to 50 tasks. Progress count shown in section header.
6 • Deliverables	Track deliverables (documents, sign-offs, artifacts) with the same checklist-style UI as Tasks. Up to 50 deliverables. Progress count shown in header. Deliverables appear in schedule export and print report.
7 • Notes	Free-text activity notes field (up to 2,000 characters). Violet dot on section header when notes are present. Notes are for reference only and do not affect scheduling or simulation.
8 • Schedule Analysis	Dependency Mode only. Read-only. Shows Total Float and Free Float for this activity. Activities on the critical path display Total Float = 0.

SCHEDULING CONSTRAINTS (DEPENDENCY MODE)

Scheduling constraints impose date boundaries on an activity's start or finish. Apply them via the Activity Edit modal → Scheduling Constraint section, or by clicking the constraint badge in the activity grid. Hard constraints adjust the schedule and are enforced in every Monte Carlo trial. Soft constraints flag violations without forcing the schedule.

Type	Full Name	Description
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MSO	Must Start On	Activity must start on exactly the specified date.
MFO	Must Finish On	Activity must finish on exactly the specified date.
SNET	Start No Earlier Than	Activity cannot start before the specified date.
SNLT	Start No Later Than	Activity cannot start after the specified date.
FNET	Finish No Earlier Than	Activity cannot finish before the specified date.
FNLT	Finish No Later Than	Activity cannot finish after the specified date.
Hard mode	The scheduler adjusts start/finish dates to satisfy the constraint. Each Monte Carlo trial also respects the constraint — buffer calculations reflect the constrained schedule.	
Soft mode	The constraint date is advisory only. Violations are flagged in the Warnings Panel with an amber indicator, but the schedule is not adjusted.	
Warnings Panel	Collapsible panel below the activity grid listing all constraint conflicts (hard — red) and soft violations (amber) with plain-language explanations.	

SCENARIO FEATURES

Baseline scenario	Every project starts with a Baseline scenario representing your original plan. Clone it to model alternatives. Any scenario can be deleted — the last remaining scenario in a project is the only one protected from deletion.
Scenario tabs	Clone any scenario to create alternatives (e.g., optimistic, pessimistic). Scenario tabs scroll horizontally when more scenarios exist than fit in the view — the Compare button stays pinned on the right edge. Drag the grip handle on any scenario tab to reorder tabs. Lock/unlock a scenario to prevent accidental edits.
Finish Target date	Set a project-level target finish date (e.g., the committed date from your project charter) directly in the Scenario Summary Card. A dashed vertical line marks the target on the Gantt chart (toggle via Finish Target in the Gantt toolbar — the chart timeline extends automatically if the target date falls past the project buffer end).
RAG schedule health	After simulation, the Finish Target date and Gantt line are color-coded: Green = simulation finishes by the green percentile threshold (default P80), Amber = within the amber threshold (default P50), Red = at risk. Thresholds are configurable in Settings → Finish Target — Schedule Health.
Scenario Notes	Free-text notes field per scenario (up to 2,000 characters). Access via the memo icon in the Scenario Summary Card. A blue dot on the icon indicates notes are present.
Parkinson's Law toggle	Per-scenario toggle (default: on) in the Scenario Summary Card. When on, no simulated duration falls below the deterministic schedule. Disable to model optimistic scenarios where activities may finish early. An amber note appears when disabled. The default is configurable in Settings → Preferences.
Dependency Mode	Enable to add Finish-to-Start / Start-to-Start / Finish-to-Finish relationships with optional lag, apply scheduling constraints (Hard/Soft), compute CPM critical path (highlighted in Gantt), and surface Total Float and Free Float per activity. Float appears in the Activity Edit modal → Schedule Analysis, in Gantt hover tooltips, and in schedule exports.
Milestones	Dependency Mode only. Named milestones with optional target dates. Assign activities to milestones; the simulation computes buffer days per milestone. Milestone markers appear on the Gantt chart.

Scenario comparison	Compare 2–3 scenarios side-by-side: dates, duration, buffer, and a CDF overlay chart. Copy-image buttons on the metrics table and CDF chart let you paste a clean PNG directly into slides or documents (Firefox-aware: disabled with tooltip on unsupported browsers). Access from the scenario toolbar.
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GANTT CHART

Views	Two views: Deterministic (scheduled dates) and With Uncertainty (bars extended to a Monte Carlo percentile). Switch via the toolbar.
Appearance panel	Click the palette icon in the Gantt toolbar to open the collapsible Appearance Panel. Controls: Name Column (Narrow / Normal / Wide), Font Size (Small / Normal / Large / XL), Row Size (Compact / Normal / Comfortable), Bar Label (Duration / Dates / None — auto-hidden when bar is too narrow), Timeline Labels (Sparse / Normal / Dense), Fit to Window checkbox, Non-work day shading (gray bands for weekends and holidays), and Row guide lines (faint horizontal lines every 3 rows — on by default). All settings are saved with the project and survive export/import round-trips. Print Gantt mirrors all appearance settings.
Color presets	Ten built-in themes: Classic (default blue/orange), Professional, Colorful, Grayscale, Contrast, Forest, Ocean, Sunset, Lavender, and Earth — each with light and dark mode variants. Override Planned, In Progress, or Completed bar colors with any custom color via the swatch picker or native color input. Custom colors clear when switching presets.
Fit to Window	When enabled, compresses the full project timeline into the visible container width — no horizontal scrolling. Essential for multi-year programs: the complete Gantt fits on screen and the Copy as PNG button captures the entire chart in one image, ready to paste into PowerPoint or Outlook.
Timeline Labels (multi-year)	For projects spanning more than 18 months (>540 days), controls tick density: Dense — monthly ticks (Apr '26, May...) · Normal — quarterly ticks (Q1 '26, Q2, Q3, Q4, Q1 '27...) · Sparse — semi-annual ticks (H1 '26, H2, H1 '27...). Year labels appear on the first tick and at every year boundary (shown in bold). For projects ≤18 months, ticks auto-select daily / weekly / biweekly / monthly based on span.
Interactive actions	Click a Gantt bar → Activity Edit modal. Click a dependency arrow (Dependency Mode) → Dependency Edit modal (change type, lag, or delete). Click an activity name → rename inline. Hover arrows for visual highlighting. Terminal markers (right-edge stripe) identify activities with no successor. In sequential mode (Dependency Mode off), the Gantt auto-draws non-interactive finish-to-start arrows to make the implicit ordering visually explicit. Section headers appear as bold labels with a full-width rule across the chart; click the label to rename inline (disabled when the scenario is locked).
Toolbar toggles	Today line · Project name · Show Activity IDs (#1, #2... — persisted per project) · Finish Target (dashed RAG line — always visible when toggled on, extending the chart timeline if the target date falls past the project buffer end; disabled with tooltip when no target date is set) · Dependency arrows show/hide (available in both modes — hides auto-drawn arrows in sequential mode) · Critical path highlight · Palette (appearance panel).
Copy & print	Copy the Gantt as a PNG to clipboard (disabled with tooltip in Firefox). Enable Fit to Window first to capture the complete timeline for long projects. Print the full project report — Gantt mirrors all appearance settings including layout dimensions, bar colors, font scaling, weekend shading, and bar labels. A 'Date prepared' label appears in the bottom-right corner.

■ **Tip:** For a presentation-ready multi-year Gantt: enable Fit to Window, set Timeline Labels to Normal (quarterly) or Sparse (semi-annual) depending on project length, then click Copy as PNG. The complete chart — all activities across the full timeline — copies to clipboard in one shot. The Appearance Panel settings are stored on the project, not globally — each project can have its own layout and theme.

CALENDAR, WORK WEEK & HOLIDAY LOADER

Work week	Click interactive day pills to toggle work days on/off — any combination including non-contiguous schedules (e.g., Mon/Wed/Fri). A warning appears if the configuration produces no valid work days.
Country holiday loader	Load public holidays from 100+ countries via the Nager.Date API. Multi-country support: holidays from multiple countries are merged additively. Built-in US holidays serve as an offline fallback.
Manual holidays	Enter custom non-work dates (e.g., company shutdowns, regional observances) with an optional locale label. Smart validation warns if a date is already a work day or falls on an API holiday.
Converted work days	Override specific non-work days as work days on a per-project basis (e.g., make a specific Saturday count). An amber warning appears if the list exceeds 50 entries.
Project vs. global	Calendar settings (work week, converted days) are per-project. Holiday lists are global across all projects. All scheduling, Gantt charts, and exports respect the active calendar configuration.

DATA, STORAGE & EXPORTS

Local storage	All project data is stored in your browser's localStorage by default — nothing leaves your device. No analytics, no telemetry, no third-party data collection. An amber caution banner appears on every page while data is stored locally, reminding you to export before closing. Dismiss with Got it for the session, or suppress it permanently via Settings → Notifications. Clearing browser site data permanently deletes local-only projects.
Cloud storage	Enable by clicking the auth chip in the app header to open the Storage & Sign In modal, or navigate to Settings → Cloud Storage. Sign in with Google or Microsoft. Cloud mode syncs projects to Firestore across all your devices. Share projects with teammates from Settings → Sharing: add collaborators by email individually, or paste multiple email addresses at once for bulk invitations — existing SPERT Scheduler users are added immediately; new users receive an invitation email and claim the project on sign-in. Owner, editor, and viewer roles are supported. Requires acceptance of the Terms of Service.
Export / Import (JSON)	Export any single project as a JSON file for backup or transfer. Export All Projects (Projects page header) exports every active project in one click. Import Projects (Projects page header) restores one or more projects from JSON — including all scenarios, activities, constraints, dependencies, tasks, deliverables, notes, section headers, and Gantt appearance settings. Import detects both ID and name conflicts, each with per-conflict resolution (skip / replace / copy — ID conflicts default to skip, name conflicts to copy). An opt-in preferences import toggle (default: off) applies imported preference settings.
Import from spreadsheet	Settings → Import Activities from Spreadsheet. Accepts a CSV file or clipboard paste. Download the template for format guidance. Live preview with row-level error/warning display. Import as a new project or add as a new scenario to an existing project.
Schedule export (XLSX/CSV)	Download the activity schedule from the Scenario Summary Card buffer row. XLSX: bold headers, frozen row, auto-width columns, metadata summary block. Dependency Mode adds Predecessor/Successor columns (e.g., 1FS+2d, 2SS, 3FF), constraint columns (type, date, mode, note), float columns (Total Float, Free Float), and Tasks and Deliverables columns (summary + details). Section headers appear as labeled rows in the export — they are skipped on CSV/Excel re-import without error. Export to JSON to preserve section headers across round-trips.

Simulation CSV export	Export raw Monte Carlo trial results from the Simulation Results panel. One row per trial — useful for external analysis or audit.
Preferences	Settings → Preferences: default distribution, confidence level, activity and project probability targets, trial count, heuristic percentages, Parkinson's Law default, date format, and RAG schedule health thresholds. Settings → Export Schedule: configure schedule export options. Settings → Notifications: suppress the localStorage caution banner permanently.

Note: Simulation runs automatically after most estimate changes. If you need to force a re-run (e.g., after bulk-editing or changing calendar settings), click Run Simulation in the Scenario Summary Card.

SIMULATION RESULTS & CHARTS

Histogram	Project duration distribution across all Monte Carlo trials. Buffer shading highlights the range between the deterministic finish and your project target percentile. Parkinson's Law clamping is reflected in the distribution shape.
CDF chart	Cumulative distribution function. Read directly: 'There is an X% probability of finishing within Y days.' Hovering over the curve shows the projected calendar finish date alongside duration and probability. Enter a target finish date in the field above the chart to see the probability of finishing by that date, with a RAG-colored reference line (green/amber/red based on your Schedule Health thresholds). The project target percentile is marked. Scenario comparison CDF tooltips also include projected finish dates.
Percentile table	P5 through P99 in working days, with corresponding calendar finish dates. Use to select a project target that matches your risk tolerance.
Bootstrap CI	Optional 95% confidence interval on each percentile estimate. Enable via the toggle in the Simulation Results panel. Shows simulation precision.
Sensitivity analysis	Ranks activities by their variance contribution to the project total. Identifies which estimates have the greatest impact on schedule risk. Available for projects with 2+ activities.
Scenario comparison	Compare 2–3 scenarios side-by-side (dates, duration, buffer) plus a CDF overlay chart for visual comparison. Access from the scenario toolbar.

■ **Tip:** Press ? anywhere in the app to open the keyboard shortcuts panel. Common shortcuts: R = Run Simulation, D = toggle Dependency Mode, G = toggle Gantt view.

TIPS & BEST PRACTICES

Activity status workflow	Mark activities In Progress as work begins — Actual Duration auto-fills with elapsed working days and is editable to override. Mark Complete and confirm or adjust Actual Duration (and optionally Actual Finish Date) in the Activity Edit Modal → General section. The simulation uses actuals for completed activities and models remaining uncertainty for in-progress ones.
Parkinson's Law toggle	Keep Parkinson's Law on for baseline planning (work expands to fill available time). Turn it off to model an optimistic scenario where activities may finish ahead of schedule. Compare the two scenarios to understand the planning assumption's impact.
Constraint best practices	Use Hard constraints only for genuine external date commitments (regulatory deadlines, vendor delivery dates). Use Soft constraints for aspirational targets. Review the Warnings Panel after adding constraints — hard conflicts must be resolved before the schedule is valid.

Tasks & Deliverables	Use the Tasks section for step-by-step work within an activity (up to 50 items). Use the Deliverables section for tangible outputs and sign-offs (up to 50 items). Both appear in the schedule XLSX/CSV export and in the printed report. Progress bars in the activity grid give a quick visual status at a glance.
Activity Notes & Scenario Notes	Record assumptions, risks, or review comments in activity Notes (up to 2,000 chars) via the Activity Edit modal. Use Scenario Notes (memo icon in summary card) for scenario-level context — e.g., 'This scenario assumes vendor delivers by April 15.' A violet indicator beneath the activity name and a violet dot on the Notes section header show when activity notes are present. A blue dot on the Scenario Notes icon indicates scenario notes are present.
Bulk activity actions	Select activities via checkboxes to reveal the Bulk toolbar. Stage a Status (Planned / In Progress / Complete), Confidence level, and/or Distribution type change, then click Apply to apply all at once. Bulk-marking Complete auto-fills Actual durations from elapsed working days.
Gantt Appearance	Use the palette icon to customize the Gantt for each project. For multi-year programs: enable Fit to Window and set Timeline Labels to Normal (quarterly) or Sparse (semi-annual), then use Copy as PNG for a complete, scroll-free presentation image. Settings travel with the project — appearance is preserved when you export and re-import.
Interactive Gantt workflow	Click Gantt bars to edit any activity without scrolling back to the grid. Click dependency arrows to adjust relationship type or lag. Use terminal markers (right-edge stripe) to quickly spot activities with no successors — these are candidates for float analysis or milestone assignment.
Finish Target & RAG	Set a Finish Target date in the summary card to activate RAG health tracking. Green = simulation P80 (or your configured threshold) finishes on or before the target. Amber = P50 meets the target. Red = at risk. Adjust the thresholds in Settings → Finish Target — Schedule Health to match your organization's risk appetite.
Scenario workflow	Start with the Baseline, then clone it to explore alternatives. Use Scenario Comparison to present the trade-offs. Drag scenario tabs to put them in a logical order. Lock scenarios you've approved to prevent accidental edits.
Sensitivity analysis	Run Sensitivity Analysis after building your full schedule to identify high-variance activities. Prioritize those for tighter estimation, additional risk mitigation, or closer tracking during execution.
Section headers	Use the + Section button to group related activities visually. Assign a color for a tinted section header row in the grid; section headers appear in the Gantt as bold labels with a full-width rule, in printed reports, and in CSV/Excel exports (labeled rows — not reconstructed on re-import). Export to JSON to preserve section headers across round-trips.
Data safety	Use Export All Projects (Projects page header) to back up all active projects at once. The amber localStorage banner reminds you to export before closing — suppress it permanently in Settings → Notifications once you've established a backup routine. Use Cloud Storage for automatic cross-device backup and team sharing.

■ **Tip:** The Schedule XLSX export is the most complete artifact from SPERT® Scheduler — it includes activity dates, float values, tasks, deliverables, constraint details, and dependency relationships. Share it with stakeholders who do not have access to the app.